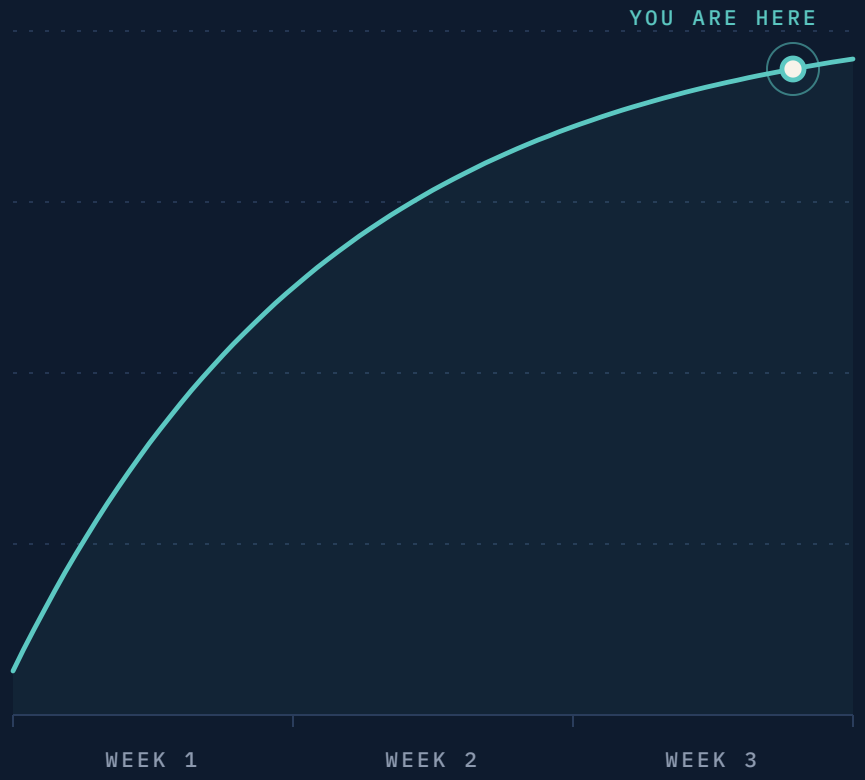
WEEK 3 · DAY 4 · OVERVIEW

PCA and t-SNE



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Course 3 described the curve with measures you chose: the **2s10s** slope and the **2s5s10s** fly.

Today the file chooses.

A desk wants eleven Treasury yields' own history, 2015 to 2026, to say how many separate ways the curve moved and what shape each one has.

It also wants one picture of many bond columns at once.

The Treasury file is real, and everything below describes its history, never what rates will do.

The bonds are invented for this course.

OBJECTIVES

Find the curve's shapes with PCA and read them, rebuild curves from three numbers, keep PCA inside a pipeline, scale before PCA, and draw the bonds with t-SNE.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 4

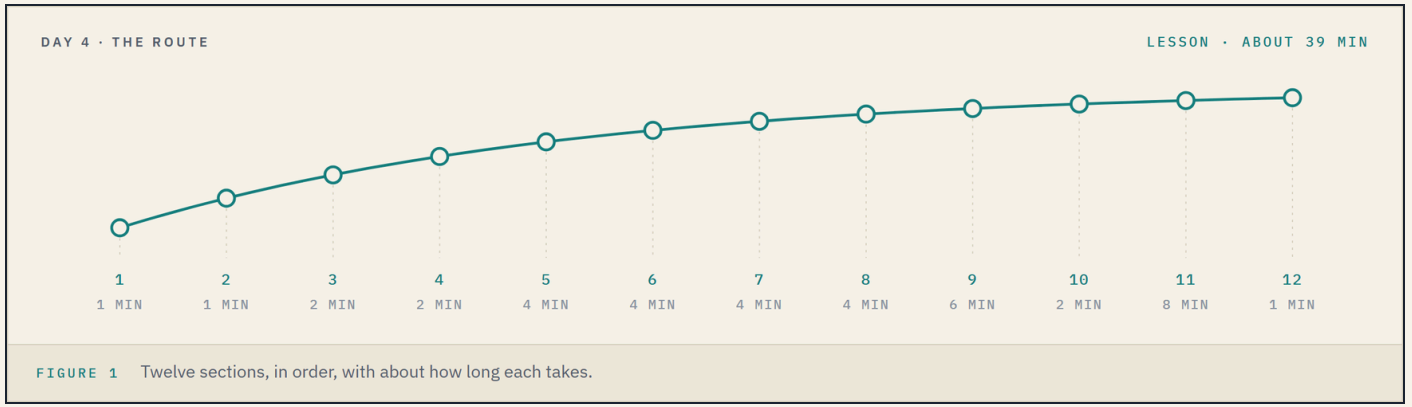


FIGURE 1 Twelve sections, in order, with about how long each takes.

	SECTION	ABOUT
1	Setting Up	1 min
2	Loading the Data	1 min
3	Data Overview	2 min
4	Exploratory Data Analysis	2 min
5	PCA in Two Columns	4 min
6	PCA on the Curve Levels	4 min

	SECTION	ABOUT
7	Three Numbers a Day	4 min
8	PCA on Daily Changes	4 min
9	PCA Inside a Pipeline	6 min
10	Scaling Before PCA	2 min
11	t-SNE: A Picture of Neighbours	8 min
12	Insights and Conclusion	1 min

Time. About 39 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

THE SEED AND THE THREAD SETTINGS

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
```

WHERE THE DATA LIVES

```
DATA = "../../../data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from ipywidgets import interact
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE, trustworthiness
from scipy.spatial.distance import pdist
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.model_selection import TimeSeriesSplit, cross_val_score
from sklearn.linear_model import LinearRegression
from sklearn.cluster import KMeans
sns.set_theme(style="whitegrid")
sns.set_palette("colorblind")
plt.rcParams["figure.figsize"] = (10, 5)
```

Run this cell and move on: it fixes the seed and finds the course data.

Read the Treasury curve and the 821 bonds.

Loading the Data

THE FILE

```
curve = pd.read_csv(DATA + "treasury_par_yields.csv", parse_dates=["date"], index_col="date")
bonds = pd.read_csv(DATA + "benchmark.csv")
ratings = pd.read_csv(DATA + "ratings.csv")
bonds = bonds.merge(ratings[["rating", "score"]], on="rating")
bonds["log_oas"] = np.log(bonds["oas"])
```

THE DATA DICTIONARY

- **`date`**: the market date
- **`1 Mo`** to **`30 Yr`**: eleven par yields with no gaps, in percent
- **`1.5 Month`**, **`2 Mo`**, **`4 Mo`**: tenors that start partway through the file, not used
- **`rating`** and **`score`**: the credit rating, and its score from 1 (AAA) to 18 (CCC)
- **`oas`** (bp), **`duration`** (years), **`coupon`** (percent)

Data Overview

DATES, TENORS AND GAPS

```
(2940, 14) 2015-01-02 2026-10-02
1 Mo            0
1.5 Month    2532
2 Mo          949
3 Mo            0
4 Mo         1951
6 Mo            0
1 Yr            0
2 Yr            0
3 Yr            0
5 Yr            0
7 Yr            0
10 Yr           0
20 Yr           0
30 Yr           0
dtype: int64
```

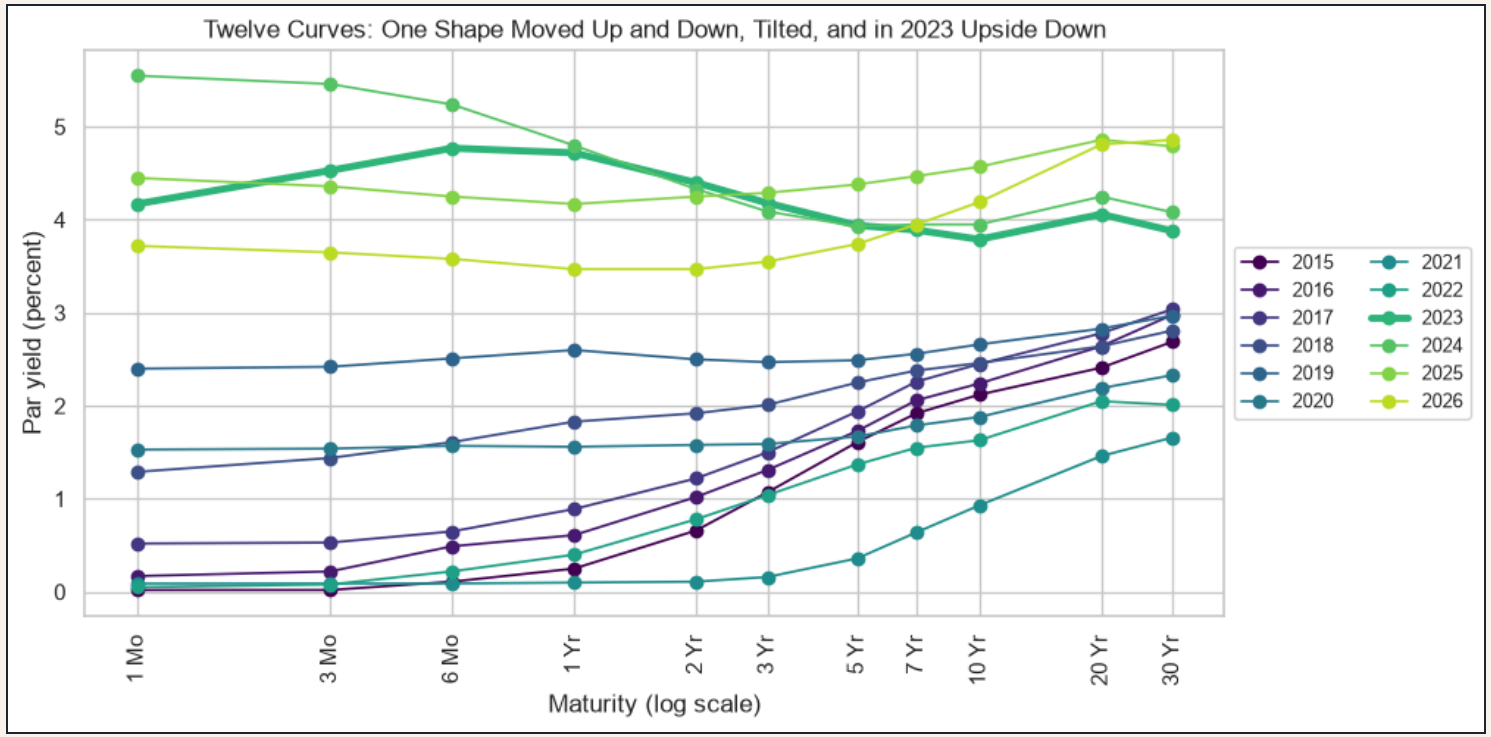
THE ELEVEN CLEAN TENORS

	MEAN	STD
1 Mo	2.10	1.98
3 Mo	2.16	1.96
6 Mo	2.22	1.91
1 Yr	2.24	1.79
2 Yr	2.28	1.60
3 Yr	2.34	1.48
5 Yr	2.47	1.32
7 Yr	2.64	1.24
10 Yr	2.76	1.19
20 Yr	3.13	1.17
30 Yr	3.22	1.06

Eleven clean columns in one unit, percent, with the short end the most restless.

Draw the curve on each year's first day.

Exploratory Data Analysis



THREE TENORS ON THE TWELVE DATES

DATE	3 MO	10 YR	30 YR
2015-01-02	0.02	2.12	2.69
2016-01-04	0.22	2.24	2.98
2017-01-03	0.53	2.45	3.04
2018-01-02	1.44	2.46	2.81
2019-01-02	2.42	2.66	2.97
2020-01-02	1.54	1.88	2.33
2021-01-04	0.09	0.93	1.66
2022-01-03	0.08	1.63	2.01
2023-01-03	4.53	3.79	3.88
2024-01-02	5.46	3.95	4.08
2025-01-02	4.36	4.57	4.79

Twelve curves, each mostly one curve moved up or down, with a tilt: we ask how many ways of moving there really are.

PCA in Two Columns

One date's PC1 score by hand: 2015-01-02, two tenors

	2 Yr	10 Yr
yield on the date (percent)	0.66	2.12
minus the tenor's average	- 2.28	- 2.76
= centred yield	-1.62	-0.64
times the PC1 loading	x 0.809	x 0.588
= weighted	-1.31	-0.38

→ add them up **-1.69** the PC1 score

averages over 2,940 dates; loadings from PCA on the two tenors, signed so they sum to a positive number

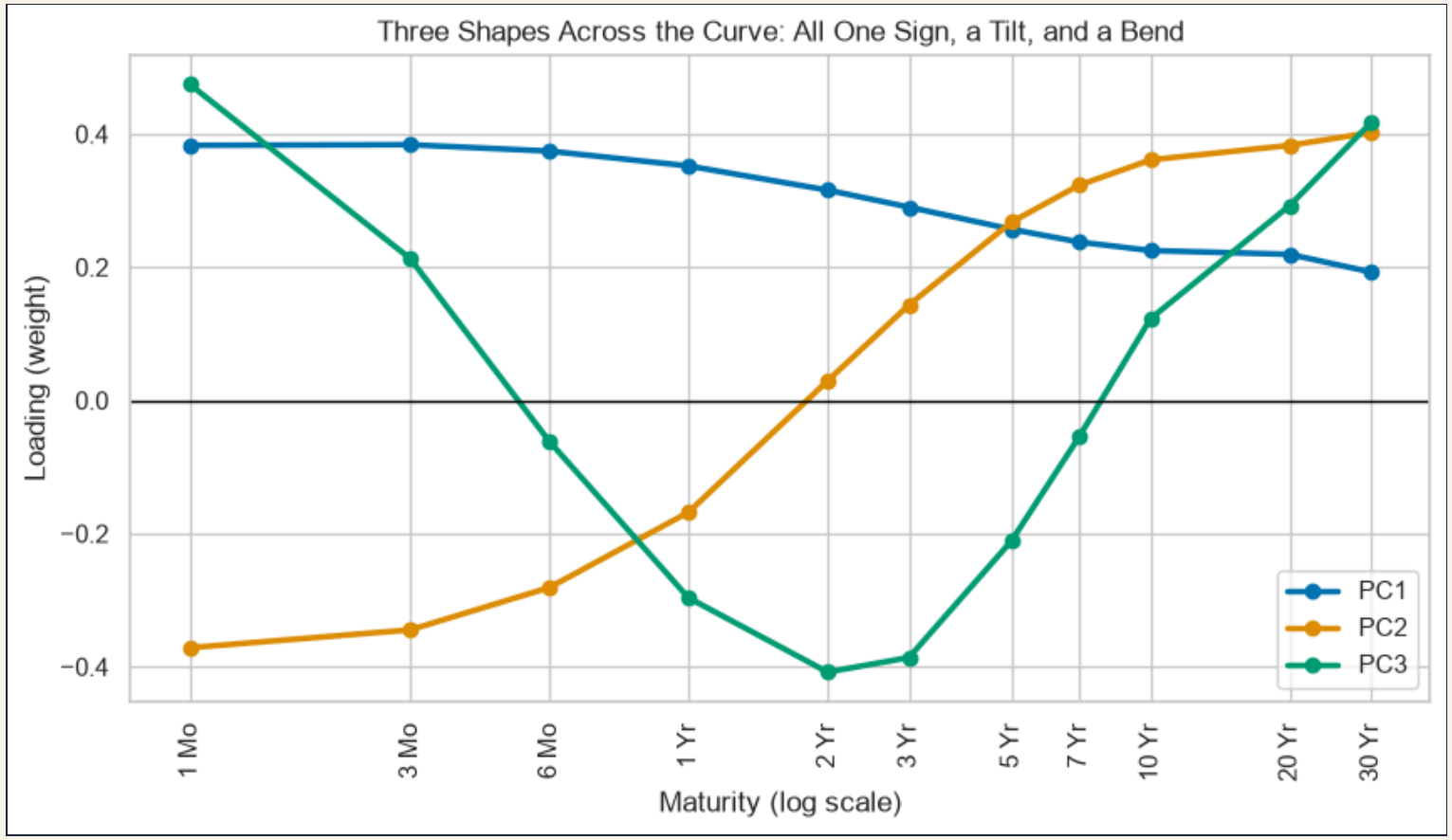
FIGURE 2 The first date's PC1 score by hand: centred yields -1.62 and -0.64 times 0.809 and 0.588 add up to -1.69.

WHAT PCA ON TWO TENORS PRINTS

```
Correlation: 0.947
Averages (percent): [2.28 2.76]
Share of the variance (percent): [97.57  2.43]
Loadings, one row per component:
[[ 0.809  0.588]
 [-0.588  0.809]]
```

The tenors correlate 0.947, and PC1 carries 97.57 percent of the variance, loadings 0.809 and 0.588.

PCA on the Curve Levels



READING THE LOADINGS

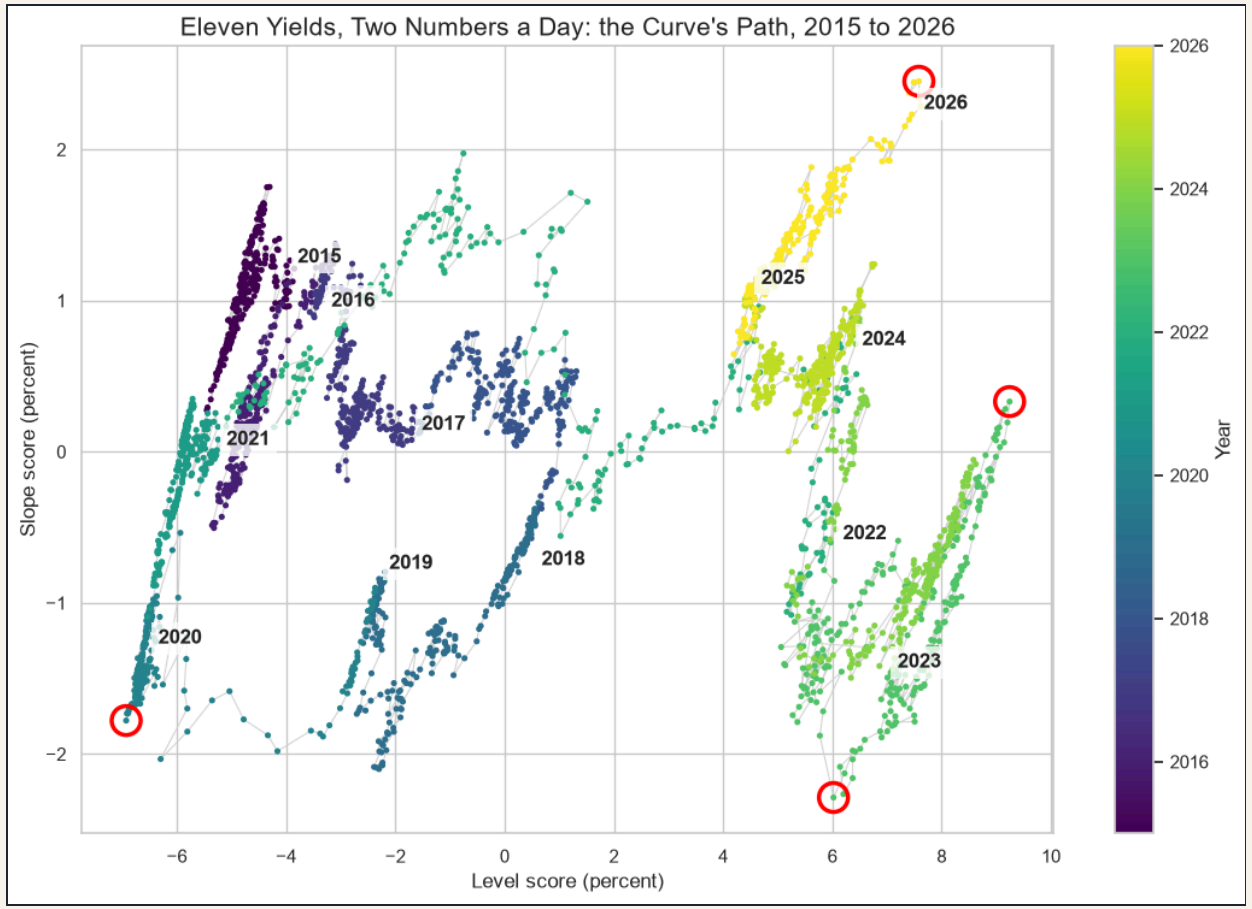
PC1 is positive on every tenor, 0.193 to 0.384: every yield moves one way.

PC2 is negative to the 1 Yr and positive from the 2 Yr: one end against the other.

PC3 is positive at both ends and negative from the 6 Mo to the 7 Yr: belly against wings.

These name shapes of weights in this file's history, not what moves rates.

Three Numbers a Day



AVERAGE MISS WITH K SCORES KEPT

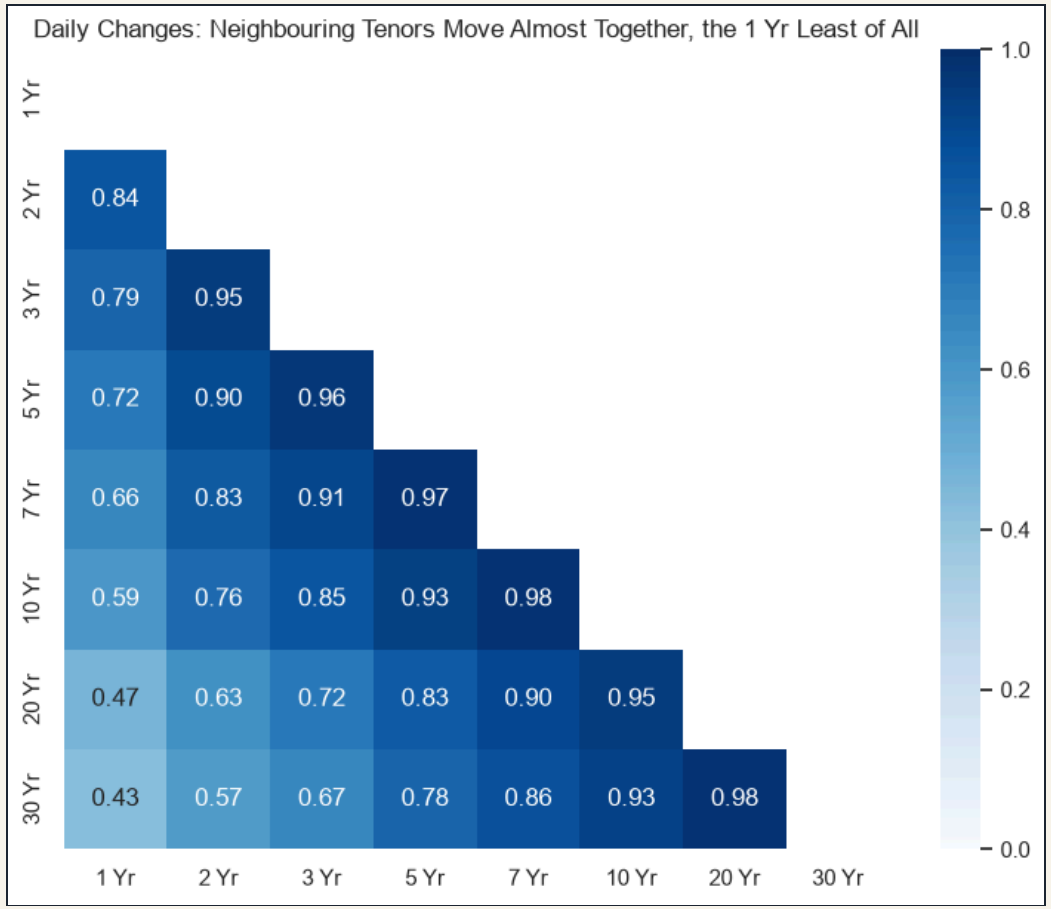
```
1 components: average miss 25.84 bp
2 components: average miss 10.6 bp
3 components: average miss 4.35 bp
4 components: average miss 2.85 bp
```

THE PATH

The path runs left to the lowest level on 2020-08-04, then right to the highest on 2023-10-19.

Over the span, three shapes carry the curve. A desk asks about days, so we repeat it on daily changes.

PCA on Daily Changes



EACH DAY'S CHANGE IN BP, EIGHT TENORS

```
tenors = ["1 Yr", "2 Yr", "3 Yr", "5 Yr", "7 Yr", "10 Yr", "20 Yr", "30 Yr"]
changes = levels[tenors].diff().dropna() * 100
print(changes.shape)
```

```
(2939, 8)
```

READING THE DARKEST AND THE PALEST CELL

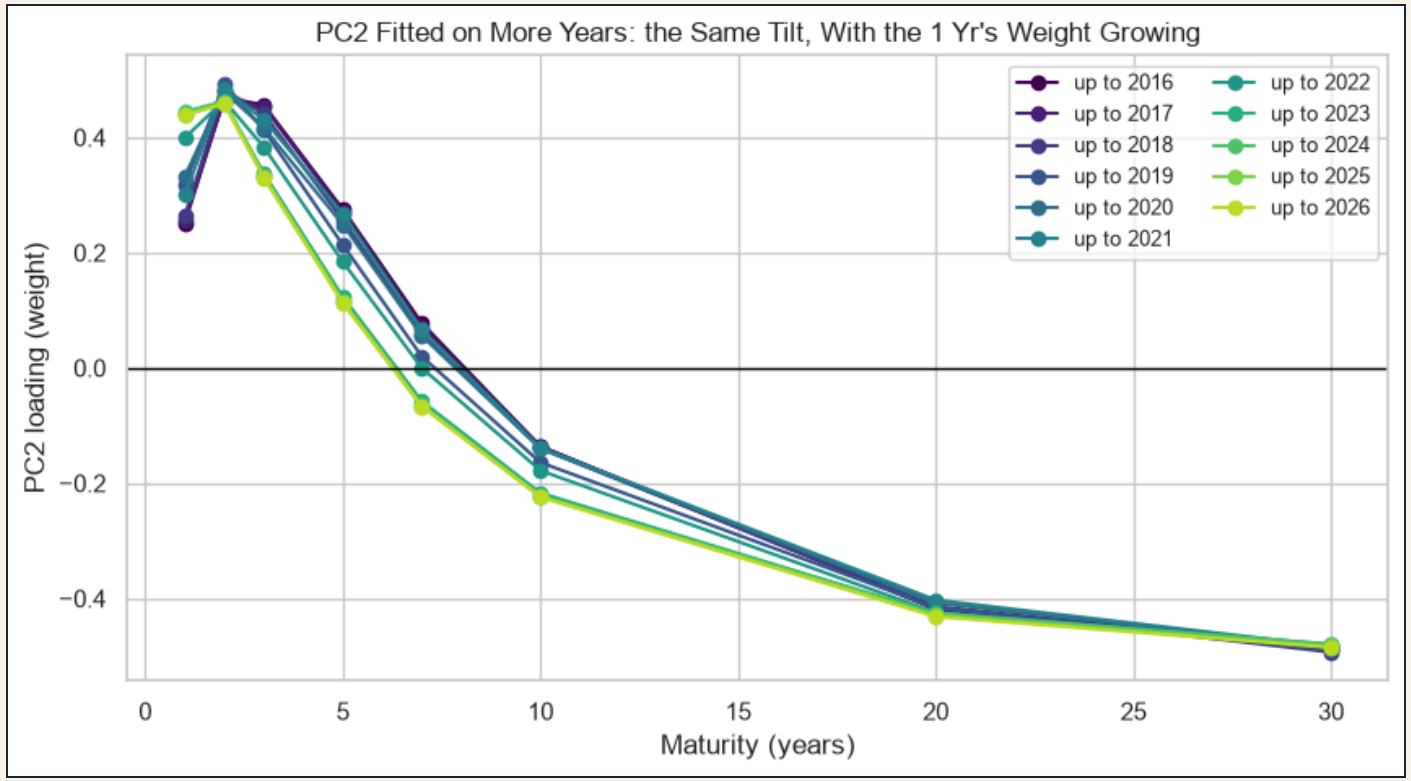
The 20 Yr and 30 Yr changes correlate 0.98; the 1 Yr and 30 Yr, the palest cell, 0.43.

THE QUESTION

Q. On the levels, PC3 was curvature. Sketch PC3 of the daily changes across the eight tenors.

From here we keep the eight tenors from the 1 Yr out, where Course 3's slopes and flies live.

PCA Inside a Pipeline



FOLD 1 OF THE WALK-FORWARD

```
Fold 1 trains on 545 dates, up to 2017-03-08
Averages match fold 1's training dates: True
Averages match the whole file: False
```

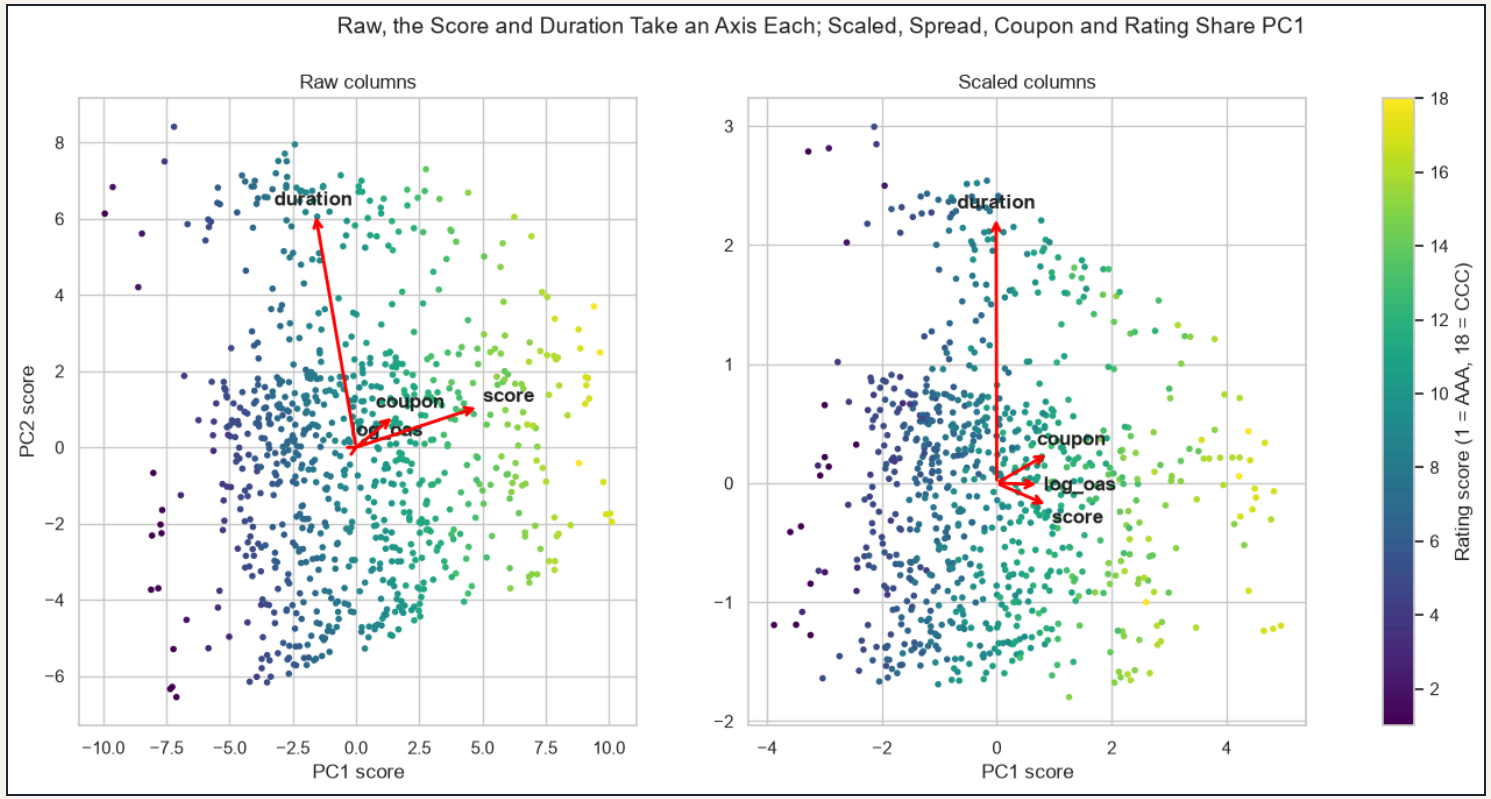
WHAT THE FITS SHOW

The tilt keeps its shape, and the 1 Yr's weight grows from 0.251 to 0.438.

So a whole-file PCA gives every date loadings that later dates helped choose: Course 3's lookahead, through a fitted step.

Compare the bonds' components raw and scaled.

Scaling Before PCA



RAW, THEN SCALED

Raw, PC1 is mostly the rating score, 0.897, and PC2 mostly duration, 0.963: the two columns with the largest standard deviations.

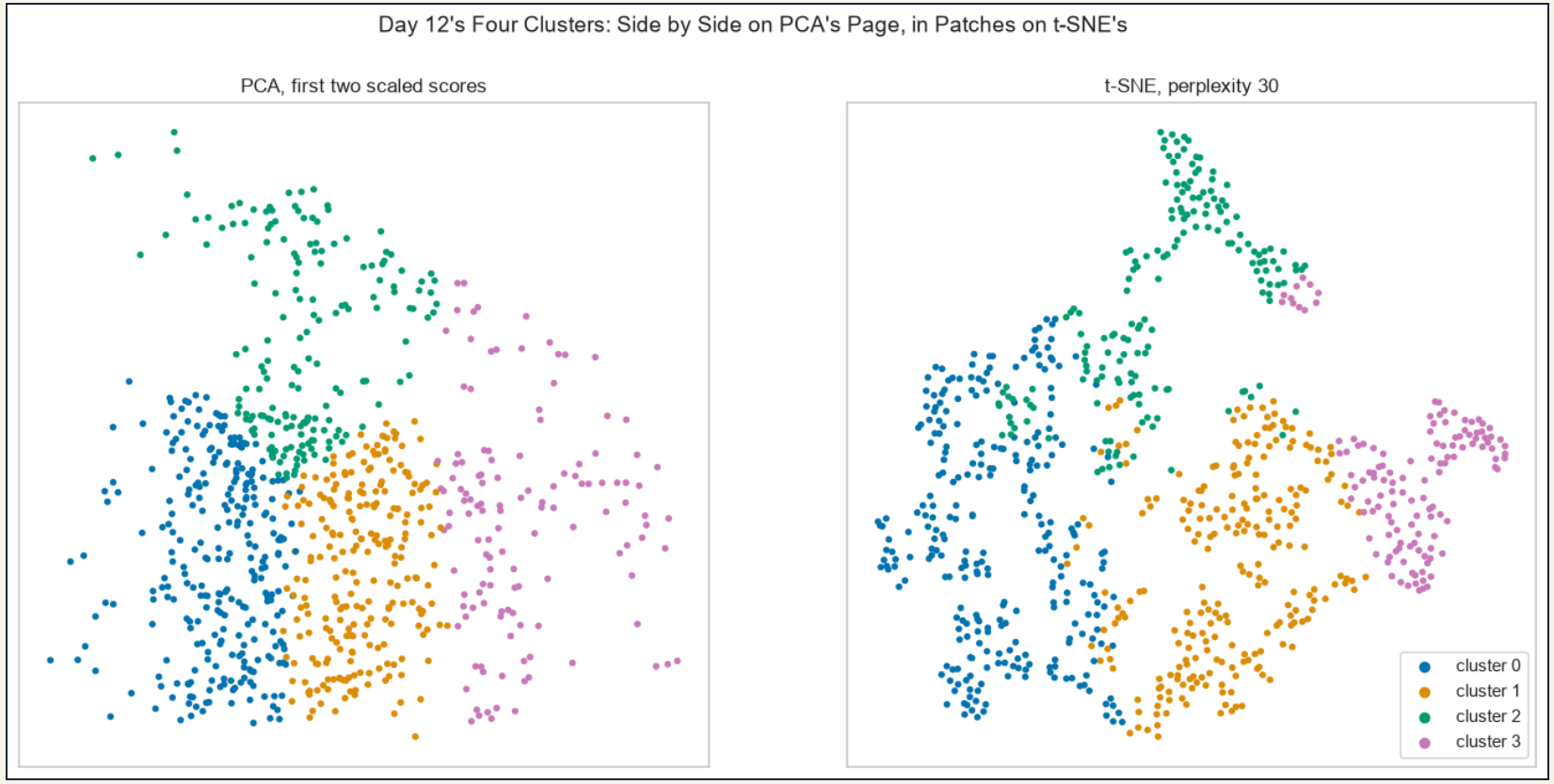
Scaled, PC1 is log OAS, coupon and score together, and PC2 is duration alone, 0.981.

DURATION IN MONTHS, NO SCALING

```
log_oas    -0.000
duration    1.000
coupon      0.004
score      -0.009
Name: PC1, dtype: float64
PC1 share: 99.03 percent
```

Day 12's rule holds for PCA: units choose the axes, so scale first. Two straight axes keep 64.96 and 25.70 percent of the scaled variance; t-SNE draws a different kind of picture.

t-SNE: A Picture of Neighbours



DAY 12'S FOUR CLUSTERS

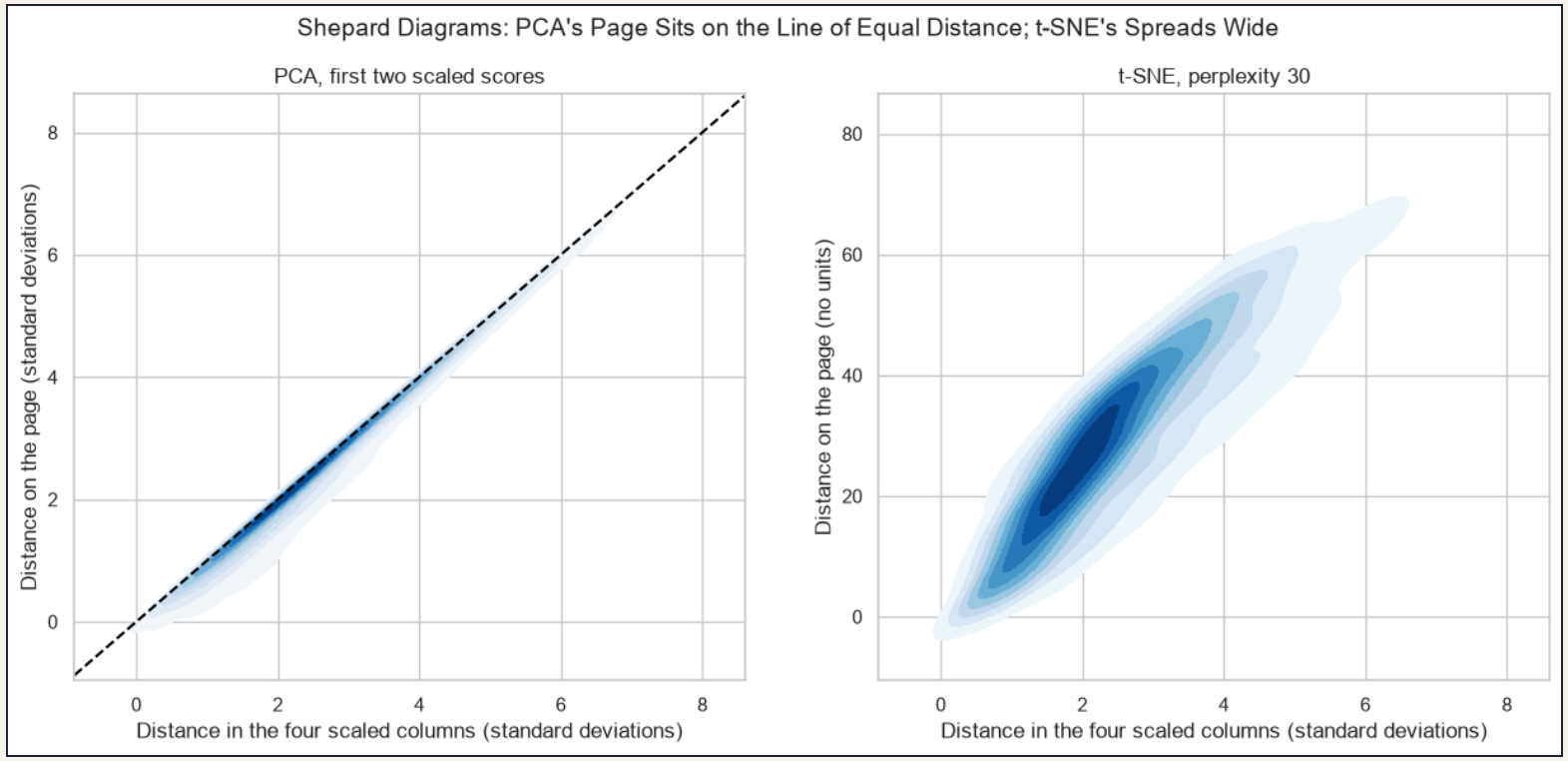
```
{0: 278, 1: 239, 2: 175, 3: 129}
```

On PCA's page the four clusters sit side by side and meet at their edges, cluster 2's long bonds at the top.

On t-SNE's page every cluster breaks into patches: cluster 2 sits in two places far apart, and a few of cluster 3's bonds sit at the tip of cluster 2's arm.

PCA keeps distances, t-SNE keeps neighbours: read a t-SNE picture for who sits near whom, never for how far or how big.

Insights and Conclusion



READING THE TWO PAGES

PCA's distances sit just under the dashed line of equal distance: a projection can only shorten a distance, and the order holds, correlation 0.986.

t-SNE's correlation is 0.879: pairs at one true distance land across a wide band on the page, so a gap between groups means little.

t-SNE keeps neighbours better than PCA and distances worse.

You Can Now

- Fit PCA, read a scree plot, and fix each component's sign.
- Name a component from its loadings, and chart its scores as history.
- Rebuild rows from k components with **`inverse_transform`**.
- Keep PCA inside a pipeline and show a step has no lookahead.
- Scale mixed units before PCA, and draw a biplot.
- Draw t-SNE pictures and judge them with trustworthiness and a Shepard diagram.

NEXT

WEEK 3 · DAY 5

Course 4 Capstone

On Day 15 the capstone puts the whole course to work on one real file.